

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0127

Elwood 230 kV
May 17, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0127. The Interconnection Customer proposes a 330.2 MW solar photovoltaic facility interconnecting at the Elwood 230 kV substation in the Illinois area. The request was received on April 24, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 84 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$25,280,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

American to Urban 345 kV L: 101.5% loading (1,270 MVA) following the outage of the East Devon to Carrel 345 kV; DFAX = 5.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
American to Urban 345 kV	Reconductor line	\$31,405,000	80% (\$25,280,000)
	Total allocated to project		\$25,280,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.